Mock Interview Edition

Autonomous Coding Agents: Planning, Execution and the Cost of a Wrong Step

16 scenarios with their follow-up probes and no answers. Use it to rehearse aloud, with a partner, or against a timer, then check yourself against the Study Edition.

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HOW THIS EDITION IS ORGANIZED

Section A, round plans: 5 ready-made sessions of five scenarios each, by theme. **Section B, question sheets:** every scenario followed by all of its probes, in order, each with its own notes box. The Study Edition page that holds the answers is printed under each scenario. **Section C, score sheet:** one row per scenario to track attempts. Question numbers and part numbers are identical in the Study Edition.

THREE WAYS TO USE IT

1. With a partner. The partner reads the scenario aloud, then asks the probes one at a time, moving on only after you have answered. **2. Alone, out loud.** Cover everything below the current probe with a sheet of paper, answer aloud for about two minutes, then slide the paper down to the next probe. **3. Alone, in writing.** Jot your key points in the notes box under each probe before looking at the next one, then compare with the Study Edition page printed under the scenario.

HOW TO GRADE A PROBE

Solid: you stated the main point, gave a number or a concrete example where one was possible, and said how you would check it. **Partial:** right direction but vague, or a number without the reasoning. **Missed:** you did not reach it, or reached the wrong conclusion. The last probe of every scenario asks for the general lesson in one sentence. Suggested pacing is about 12 minutes per scenario. Reading the answer only after you have committed to your own is what makes the practice work.

A NOTE ON THE PACING, GRADING AND ROUNDS

The 12-minute pacing, the three-level grading and the round compositions are the author's suggestions for practice, not an industry standard. The scenarios are written for this document and are not taken from a real company question bank. Where a scenario quotes a figure, the Study Edition says where it comes from.

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Every line below is a link: click it to jump to that page. The outline panel of your PDF viewer lists the same entries, and the “Contents” link in every page footer brings you back here. Question numbers are the same as in the Study Edition.

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Every scenario, by part

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03 Two bars for done Evaluation 7

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SECTION A

Round plans

Each round has five scenarios at about 12 minutes each, a 60-minute session. Mix and match freely. Click any scenario to jump to its question sheet; the number on the right is the page.

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03 Two bars for done Evaluation 7
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SECTION B

Question sheets

Each scenario is followed by its probes in order. Indentation shows how deep the follow-up goes.

PART 1

Framing the objective and reading the benchmark

Separating a multi-step agent from a single-shot generator, defining how SWE-bench grades, and reading a launch number for what it actually says.

01 Single-shot or multi-step?

FRAMING

SCENARIO

A product manager says the coding agent is just a code-completion model with a nicer prompt, so one model call per task should do. Explain why the task in this case study is a different shape.

OPENING PROBE

What does a single-shot tool do?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does this agent do instead?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Why does the difference matter for failure?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What is being optimised?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 6

02 How does SWE-bench grade?

EVALUATION

SCENARIO

A teammate says the agent 'passed' a bug because its diff looked correct on inspection. Explain how SWE-bench actually decides whether a task is resolved, and why that rule is used.

OPENING PROBE

What is in the benchmark?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How is a task scored resolved?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Why grade by tests, not by looking?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does resolve rate not capture?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 7

03 Two bars for done

EVALUATION

SCENARIO

An engineer wants to ship any patch that clears SWE-bench's grading. Explain the second, stricter bar and why the benchmark is only a proxy for it.

OPENING PROBE

What is the benchmark bar?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is the real-world bar?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Why is resolve rate only a proxy?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How should the two be used together?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 8

04 Read the 13.86% claim

EVALUATION

SCENARIO

A slide quotes: Devin resolves 13.86% of SWE-bench issues, far above the prior best of 1.96%, and 4.80% for models given the files to edit. A reviewer treats it as an independently verified result. Say what the number does and does not support.

OPENING PROBE

What are the two stated caveats?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does the subset mean in tasks?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Which way does unassisted-versus-assisted cut?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Who reported it, and what is missing?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 9

PART 2

Sizing the fleet and the cost of a task

Applying the back-of-envelope method to a fleet of long-running sessions, costing one task, and naming the input the whole estimate depends on.

05 Size the sandbox fleet

SIZING

SCENARIO

You run 500 concurrent agent sessions, each in a sandbox budgeted at 4 vCPUs, 8 GB RAM and 20 GB disk. Estimate the fleet-wide compute reserved, and say whether it is plausible.

OPENING PROBE

What is the vCPU total?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is the memory total?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Is that plausible?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Why is most of it idle?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 10

06 Cost one task

SIZING

SCENARIO

A p90 task runs 90 minutes and takes about 200 round-trips, each 3,000 input and 300 output tokens, at 3 dollars per million input and 15 dollars per million output. Sandbox compute is 0.20 dollars an hour. Estimate the cost of one task.

OPENING PROBE

What are the token totals?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is the model cost?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What is the sandbox cost?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What is the total, and which term dominates?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 11

07 Which input is most sensitive?

SIZING

SCENARIO

Your 3-dollar-per-task figure rests on 200 round-trips per task, but the source only says a task can require 'thousands of decisions'. Say which input the estimate is most sensitive to and what happens if it is wrong.

OPENING PROBE

Which input dominates the uncertainty?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What if it is ten times higher?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Why not the sandbox budget?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What should a prototype measure first?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 12

08 Cost of one benchmark run

SIZING

SCENARIO

A reviewer asks how much it costs to run the full SWE-bench once, and someone guesses 'a few dollars'. Estimate it at this document's own per-task figure, and use the result as an order-of-magnitude check.

OPENING PROBE

What is the arithmetic?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Is 'a few dollars' plausible?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Is 7 million dollars plausible?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does this not claim?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 13

PART 3

The agent core and the context it reasons over

Why the codebase is a corpus the agent keeps changing, why a long run must checkpoint, which control-flow pattern fits, and why per-step correction is not optional.

09 Why is the codebase not a static corpus?

DATA

SCENARIO

A colleague wants to index the repository once at the start of a task and reuse that index for every planning step, as a RAG chatbot indexes a document corpus. Explain why that breaks here.

OPENING PROBE

What is different from RAG?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What goes wrong with a one-time index?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does correct behaviour require?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What is the cheaper middle path?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 14

10 Why must a long run checkpoint?

DATA

SCENARIO

A 90-minute session crashes at minute 70. A colleague's design restarts it from minute 0. Explain why checkpointing is a structural requirement here, not a nice-to-have, and what the session should store.

OPENING PROBE

What does restarting from zero cost?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Why is this structural, not optional?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What state makes resume possible?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does storing the whole plan cost?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 15

11 Which control-flow pattern fits?

AGENT DESIGN

SCENARIO

An interviewer asks why the agent uses plan-and-execute with per-step observation rather than a plain ReAct loop or a single plan run to the end. Give the trade-off against both.

OPENING PROBE

Why not single-shot?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Why not plain ReAct?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Why plan-and-execute with observation?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does the hybrid cost?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 16

12 Why is per-step correction load-bearing?

AGENT DESIGN

SCENARIO

Suppose each decision is correct with probability 0.95 and a task has 50 decisions, executed open-loop with no checking in between. Estimate the chance of a clean run, and say what it implies.

OPENING PROBE

What is the formula?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is the value?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What happens at the source's scale?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does that justify?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 17

PART 4

Serving, isolation, and operating the agent

Why a long task cannot be one request, making a retried edit safe, scoping each tool's credential, bounding a sandbox escape, and catching a stalled run.

13 Why not one synchronous request?

SERVING

SCENARIO

A colleague wants the submit call to hold the HTTP connection open until the 90-minute task finishes and return the result. Explain why that cannot work and what the shape should be.

OPENING PROBE

What is the timeout problem?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is the submit shape instead?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Where does the work run?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How does the client follow progress?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 18

14 Review the retried edit action

CODE REVIEW

SCENARIO

Each agent step runs as a task-queue job retried up to max_attempts. The editor action below applies a patch. In production, a retried edit sometimes fails outright or corrupts a file. Find the faults and fix them.

```
def apply_edit(job):
    # one agent step; retried up to job["max_attempts"] on failure
    patch_path = write_patch(job["patch"])
    run(["git", "apply", patch_path], check=True)   # same path as a shell retry
    job["status"] = "succeeded"
    return job
```

OPENING PROBE

Why is a retried edit unsafe?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Why is this different from a shell retry?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What should happen before reapplying?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

State the fix.

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 19

15 Scope the tools and bound a breach

SECURITY

SCENARIO

A colleague grants the agent one 'full dev environment' credential for shell, editor and browser together. Explain the least-privilege design instead, and quantify what co-locating tenants does to a sandbox escape.

OPENING PROBE

How should the three tools be scoped?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Why scope per tool?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does co-location do to an escape?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Why does this raise the stakes beyond a basic sandbox?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 20

16 Catch a stalled run and decide the merge policy

OPERATIONS

SCENARIO

An agent edits the wrong file for an hour while every step 'succeeds' by its own account, then opens a PR that passes CI. Design the detection and say how to decide whether to merge.

OPENING PROBE

Why do timeouts alone miss this?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What detects a stall?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does a flagged session escalate to?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How to decide the merge policy?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 21

SECTION C

Score sheet

Write how many probes were solid out of the total, per attempt. Revisit anything under half after reading the Study Edition's concepts box.

No.	Scenario	Probes	Attempt 1	Attempt 2	Attempt 3
01	Single-shot or multi-step?	5			
02	How does SWE-bench grade?	5			
03	Two bars for done	5			
04	Read the 13.86% claim	5			
05	Size the sandbox fleet	5			
06	Cost one task	5			
07	Which input is most sensitive?	5			
08	Cost of one benchmark run	5			
09	Why is the codebase not a static corpus?	5			
10	Why must a long run checkpoint?	5			
11	Which control-flow pattern fits?	5			
12	Why is per-step correction load-bearing?	5			
13	Why not one synchronous request?	5			
14	Review the retried edit action	5			
15	Scope the tools and bound a breach	5			
16	Catch a stalled run and decide the merge policy	5			